



AI Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University
Lucknow



Research Seminar Series

Seminar -1

Deep Learning Based Framework for Microscopic Cell Images Classification using Multi-Level Ensemble

DATE & TIME:
6th January 2021,
4:00 PM.



SPEAKER
Ritesh Maurya

(Research Scholar, Computer
Science and Engineering,
Centre for Advanced Studies, AKTU,
Lucknow)

PLATFORM & WEB-LINK
Google Meet
(<http://meet.google.com/btc-tvde-ypc>)

ABSTRACT

With the advancement of ultra-fast microscopic image acquisition and generation techniques, there arises the need for an automated artificial intelligence (AI) based system that can complement the task of microscopic cell images classification. The earlier cell classification systems classify the cell images of a specific type captured using a specific microscopy technique, therefore, the motivation behind the present study is to develop a generic framework that can be used for the classification of cell images of multiple types captured using a variety of microscopic techniques. The proposed framework for microscopic cell images classification is based on the transfer learning-based multi-level ensemble approach. The proposed framework captures different granularities of features extracted from multiple scales of an input microscopic cell image. The weak learners used in the proposed ensemble encapsulates the aggregation of low-level coarse features and high-level semantic features, thus, represent the different granular microscopic cell image features present at a different scale of input cell images. The general applicability of the proposed framework for microscopic cell image classification has been tested with five different public datasets. The proposed framework for microscopic cell classification outperforms the other state-of-the-art classification methods in the same domain with a comparatively lesser amount of training data and consequently the shorter training time requirement.